Electric Circuit Theory

Engineering Technology

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Electric Circuit Theory (A17694)

Short Title:	Electric Circuit Theory		
Faculty	Science and Computing		
Department	Engineering Technology		
Cluster	Electronics		
Author	PJCREGG		
Credits	5	**Level:**	Intermediate

Description of Module / Aims

This course builds on the D.C. Circuit Theory and A.C. Circuit Theory modules and covers time domain and frequency domain analysis of second order electrical systems. A physical and diagrammatic approach is taken to the introduction of Maxwell's eqns as a generalisation of the basic electrical laws (e.g. Ohm's law, Kirchhoff's voltage law, Ampère's circuital law). Physical interpretations of curl, divergence etc and the surface and line integrals are emphasised.

Programmes

`ELCT-0001`	BEng (Hons) in Electronic Engineering (`WD_EONIC_B`)	2 / 4 / M
	BEng (Hons) in Information Engineering (International) (`WD_EEELC_BI`)	2 / 4 / M

Indicative Content

- Time domain: transient analysis of a 2nd order (RLC), underdamped, critical & overdamping solutions, and component estimation for given waveforms
- Frequency Domain Analysis:Transfer functions, Bode plots, pole-zero plots and root locus diagrams are developed for simple 1st & 2nd order passive (RLC) filters.
- Two Port Networks: transmission matrices are used for cascaded filters.
- Filter families: Butterworth, Chebyshev & Bessel filters - Transfer functions, Bode (amplitude & phase) plots & pole zero plots. Cauer ladder implementaions, effect of phase on pulse shape.
- Introduction to Maxwell's equations.

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Illustrate the time and frequency response of 1st and 2nd order passive electrical systems.
2. Determine equivalent circuits and component values from given responses.
3. Design passive circuits (up to 2nd order) to give desired responses.
4. Use two-port network matrices to analyse cascaded circuits.
5. Apply Maxwell's equations and constitutive equations to simple uniform field situations.
6. Perform practical experiments, use simulations and present clearly in reports.

Learning and Teaching Methods

- Lectures
- Lectorials
- Labs

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	70%	1,2,3,4,5
Continuous Assessment	30%	
In-Class Assessment	10%	1,2,3,4,5
Practical	20%	6

Assessment Criteria

70% - 100%: The learner has: attained the module learning outcomes at an excellent level; demonstrated a comprehensive knowledge of the associated subject matter; achieved an excellent level of the skills required for the subject matter; demonstrated the ability to carry out further investigation and problem solving associated with the subject matter.

60% - 69%: The learner has: attained the module learning outcomes at a very good level; demonstrated a detailed knowledge of the associated subject matter; achieved a very good level of the skills required for the subject matter.

50% - 59%: The learner has: attained the module learning outcomes at a good level; demonstrated a good knowledge of the associated subject matter; achieved a good level of the skills required for the subject matter.

40% - 49%: The learner has: attained the module learning outcomes at a basic level; demonstrated a basic knowledge of the associated subject matter; achieved a basic level of the skills required for the subject matter.

<40%: The learner has not: attained the module learning outcomes; demonstrated sufficient knowledge of the associated subject matter; demonstrated a sufficient level of the skills required for the subject matter.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Tutorial	12	
Practical	12	
Independent Learning	87	

Essential Material

- Dorf, R.C. and J.A. Svoboda. _Introduction to Electric Circuits_. 4th ed.. New York: Wiley, 1998.
- Nilsson, J.W. and S. Riedel. _Electric Circuits_. 6th ed.. New Jersey: Prentice Hall, 2000.
- Reed, M. and R. Rohrer. _Applied Introductory Circuit Analysis_. New Jersey: Prentice Hall, 1999.

Supplementary Material

- Fleisch, Daniel. _A Student's Guide to Maxwell's Equations_. Cambridge: Cambridge University Press, 2008.
- Hammond, P. and J.K. Sykulski. _Engineering Electromagnetism: Physical Processes & Computation_. Oxford: Oxford University Press, 1994.

Requested Resources

- ENGINEERING LAB: Electronics